

ENGINEERING ROADMAP • V1.0

Implementation Plan & Roadmap

Engineering Milestones, QA Matrix & Production Deployment
Playbook

PROJECT

Gymholik SaaS

VERSION

1.0.0

AUTHOR

Engineering Team

DATE

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1. Engineering Roadmap & Milestone Timeline

The Gymholik implementation roadmap spans 6 phased milestones designed for robust multi-tenant execution, zero financial discrepancy, and flawless mobile user experience.

Phase	Focus Area	Duration	Key Deliverables	Status
Phase 1	Core Multi-Tenant Engine	2 Sprints	Spring Boot 3 scaffold, MongoDB multi-tenant ThreadLocal context, JWT filter, rate limiting.	COMPLETE
Phase 2	Member & Financial Engine	2 Sprints	Membership calculations, payment ledger in paise, dues tracking, PDF receipts, reversal locks.	COMPLETE
Phase 3	Mobile Foundation & Design System	2 Sprints	Expo SDK 52, Cyber-Emerald theme tokens, custom FloatingTabBar, GlassCard, micro-interactions.	COMPLETE
Phase 4	Strict RBAC & Navigation Guards	1 Sprint	tabsByRole single source of truth, deep link guards, zero-flash role loading.	COMPLETE
Phase 5	Attendance & Personal Training	1 Sprint	QR scanning, instant manual check-in, trainer progress notes & weight progression charting.	COMPLETE
Phase 6	Production Hardening & App Store Release	1 Sprint	EAS Build profiles, Docker production staging, end-to-end multi-tenant penetration tests.	IN PROGRESS

2. Quality Assurance & Testing Matrix

Backend Automated Testing

- **Unit Tests:** JUnit 5 + Mockito covering payment calculation, membership expiry rules, and phone masking.
- **Integration Tests:** Spring MockMvc tests verifying TenantIsolationIntegrationTest and TrainerAccessIntegrationTest.

Mobile Type & Route Testing

- **TypeScript Strict Mode:** Zero compilation errors across all screens and components (npx tsc --noEmit).
- **Deep-Link Testing:** Verified redirection across 3 accounts (Owner, Staff, Trainer).

3. Production Deployment & Monitoring Architecture

1. **Backend Deployment:** Containerized via Docker, hosted on cloud container service behind Nginx reverse proxy with SSL termination and Let's Encrypt certificates.
2. **Database Cluster:** MongoDB Atlas replica set with automated point-in-time recovery and compound index monitoring.
3. **Mobile Distribution:**
 - Expo Application Services (EAS) building production binaries for iOS (TestFlight) and Android (Google Play Internal Track).
 - Sentry error tracking integrated for real-time mobile crash analysis.